Experiment 8 – Hospital Appointment Booking System

Aim:
To develop an online Hospital Appointment Booking System that allows patients to search hospitals and doctors and book appointments conveniently.

Algorithm:

1. Register and log in the patient.
2. Search hospitals by location and doctors by specialty.
3. Display available doctors and appointment slots.
4. Select the required doctor, date, and time.
5. Book the appointment and submit the check-in form.
6. Display appointment status and history.
7. Allow patients to cancel or manage appointments.
8. Allow patients to provide doctor ratings and reviews.
9. Allow admin/hospital staff to manage doctors, appointments, and hospital details.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Hospital Appointment Booking</title>

  <style>
    body {
```

```
    font-family: Arial;
    margin: 0;
    background: #eef7f9;
}

header {
    background: #00838f;
    color: white;
    text-align: center;
    padding: 20px;
}

.container {
    width: 90%;
    max-width: 850px;
    margin: 25px auto;
    background: white;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 0 10px #aaa;
}

input, select, textarea, button {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
```

```
    box-sizing: border-box;
}

button {
    background: #00838f;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background: #006064;
}

.doctor {
    border: 1px solid #ccc;
    padding: 15px;
    margin: 12px 0;
    border-radius: 8px;
}

.doctor button {
    width: 200px;
}
```

```
        .confirmation {
            background: #e0f2f1;
            padding: 20px;
            margin-top: 20px;
            border-radius: 8px;
        }
    </style>
</head>

<body>

<header>
    <h1> MedCare Hospital</h1>
    <p>Online Appointment Booking System</p>
</header>

<div class="container">

    <h2>Find a Doctor</h2>

    <input type="text" id="location"
           placeholder="Enter location">

    <select id="specialty">
        <option value="all">All Specialties</option>
        <option value="Cardiology">Cardiology</option>
```

```
    <option value="Dermatology">Dermatology</option>
    <option value="Neurology">Neurology</option>
    <option value="General">General Medicine</option>
</select>

<button onclick="searchDoctors()">
    Search Doctors
</button>

<div id="doctorList"></div>

<div id="booking" style="display:none">

    <h2>Book Appointment</h2>

    <p>
        <b>Doctor:</b>
        <span id="selectedDoctor"></span>
    </p>

    <input type="text" id="patient"
           placeholder="Patient Name">

    <input type="text" id="phone"
           placeholder="Phone Number">
```

```
    <input type="email" id="email"
           placeholder="Email">

    <label>Appointment Date</label>
    <input type="date" id="date">

    <label>Appointment Time</label>

    <select id="time">
       <option>09:00 AM</option>
       <option>10:00 AM</option>
       <option>11:00 AM</option>
       <option>02:00 PM</option>
       <option>03:00 PM</option>
       <option>04:00 PM</option>
    </select>

    <textarea id="reason"
              placeholder="Reason for appointment"></textarea>

    <button onclick="bookAppointment()">
       Confirm Appointment
    </button>

</div>
```

```
    <div id="result"></div>

</div>

<script>

const doctors = [
    {
        name: "Dr. Ravi Kumar",
        specialty: "Cardiology",
        hospital: "MedCare Hospital"
    },
    {
        name: "Dr. Priya Sharma",
        specialty: "Dermatology",
        hospital: "MedCare Hospital"
    },
    {
        name: "Dr. Arun Kumar",
        specialty: "Neurology",
        hospital: "MedCare Hospital"
    },
    {
        name: "Dr. Meena Devi",
        specialty: "General",
        hospital: "MedCare Hospital"
```

```
    }
];

let selectedDoctor = "";

function searchDoctors() {

    let specialty =
        document.getElementById("specialty").value;

    let result = doctors.filter(doctor =>
        specialty == "all" ||
        doctor.specialty == specialty
    );

    let output = "";

    if (result.length == 0) {

        output = "<p>No doctors found.</p>";

    } else {

        result.forEach(doctor => {

            output += `
```

```
        <div class="doctor">

            <h3>👨‍⚕️ ${doctor.name}</h3>

            <p>
                <b>Specialty:</b>
                ${doctor.specialty}
            </p>

            <p>
                <b>Hospital:</b>
                ${doctor.hospital}
            </p>

            <p>
                <b>Available:</b>
                Mon - Sat
            </p>

            <button onclick="selectDoctor('${doctor.name}')">
                Book Appointment
            </button>

        </div>
    `;
});
```

```
    }

    document.getElementById("doctorList")
        .innerHTML = output;
}

function selectDoctor(name) {

    selectedDoctor = name;

    document.getElementById("selectedDoctor")
        .innerText = name;

    document.getElementById("booking")
        .style.display = "block";

    window.scrollTo(0, document.body.scrollHeight);
}

function bookAppointment() {

    let patient =
        document.getElementById("patient").value;

    let phone =
        document.getElementById("phone").value;
```

```
let email =
    document.getElementById("email").value;

let date =
    document.getElementById("date").value;

let time =
    document.getElementById("time").value;

let reason =
    document.getElementById("reason").value;

if (patient == "" ||
    phone == "" ||
    email == "" ||
    date == "") {

    alert("Please fill all required details");
    return;
}

let appointmentId =
    "MED" + Math.floor(10000 + Math.random() * 90000);

document.getElementById("result").innerHTML = `
```

```
<div class="confirmation">

    <h2>✅ Appointment Confirmed</h2>

    <p>
        <b>Appointment ID:</b>
        ${appointmentId}
    </p>

    <p>
        <b>Patient:</b>
        ${patient}
    </p>

    <p>
        <b>Doctor:</b>
        ${selectedDoctor}
    </p>

    <p>
        <b>Date:</b>
        ${date}
    </p>

    <p>
```

```
            <b>Time:</b>
            ${time}
        </p>

        <p>
            <b>Reason:</b>
            ${reason || "General Consultation"}
        </p>

        <p>
            <b>Status:</b> Confirmed
        </p>

        <button onclick="window.print()">
            Print Appointment
        </button>

    </div>
  `;
}

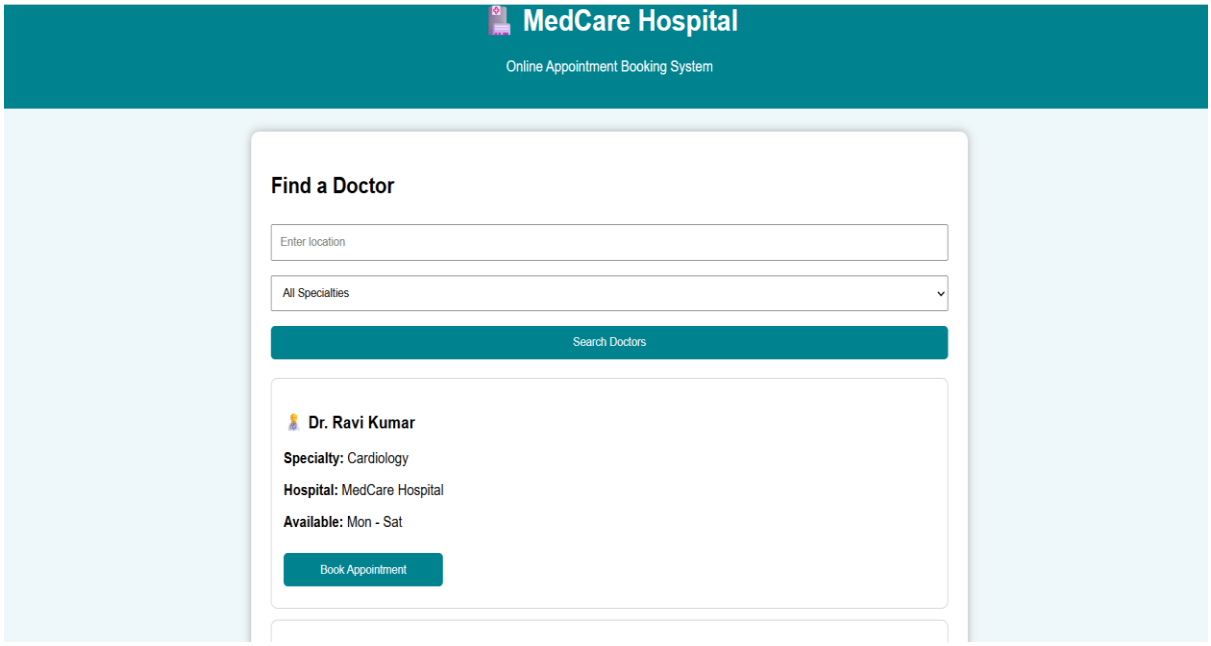
searchDoctors();

</script>

</body>
```

</html>

Output:



Result:

The Hospital Appointment Booking System was successfully developed with hospital search, doctor search, appointment booking, status tracking, reviews, and administrative management.